



3879 - Aromatics versus Aliphatics: A PAH Diagnostic Tool for JWST

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The so-called "unidentified infrared emission" (UIE) features at 3.3, 6.2, 7.7, 8.6, and 11.3 micron ubiquitously seen in a wide variety of astrophysical regions are generally attributed to polycyclic aromatic hydrocarbon (PAH) molecules. Astronomical PAHs often have an aliphatic component as revealed by the detection in many UIE sources of the aliphatic C-H stretching feature at 3.4 micron, and the aliphatic C-H deformation bands at 6.85 and 7.25 micron.

We propose a theoretical study which will facilitate the JWST community to quantitatively determine from JWST data the aliphatic fractions of the UIE carriers --- the ratio of the number of C atoms in aliphatic units to that in aromatic rings.

We will calculate the infrared emission spectra of both neutral and ionized PAHs of various sizes and a range of aliphatic contents illuminated by starlight of various spectral shapes. We will develop a "library" of PAH emission spectra and aliphatic-to-aromatic band-ratios of $I_{3.4}/I_{3.3}$, $I_{6.85}/I_{6.2}$ and $I_{7.25}/I_{6.2}$ which can be used by the astronomical community as a quantitative diagnostic tool for determining the aliphatic fractions of PAHs. We will generate a comprehensive series of $I_{3.4}/I_{3.3}$ vs. $I_{6.85}/I_{6.2}$ diagrams and $I_{6.85}/I_{6.2}$ vs. $I_{7.25}/I_{6.2}$ diagrams. These diagrams will allow us to quantitatively determine the aliphatic fraction of PAHs simply by placing the observationally-obtained band ratios in these diagrams.

OBSERVING DESCRIPTION

This is a theory proposal.